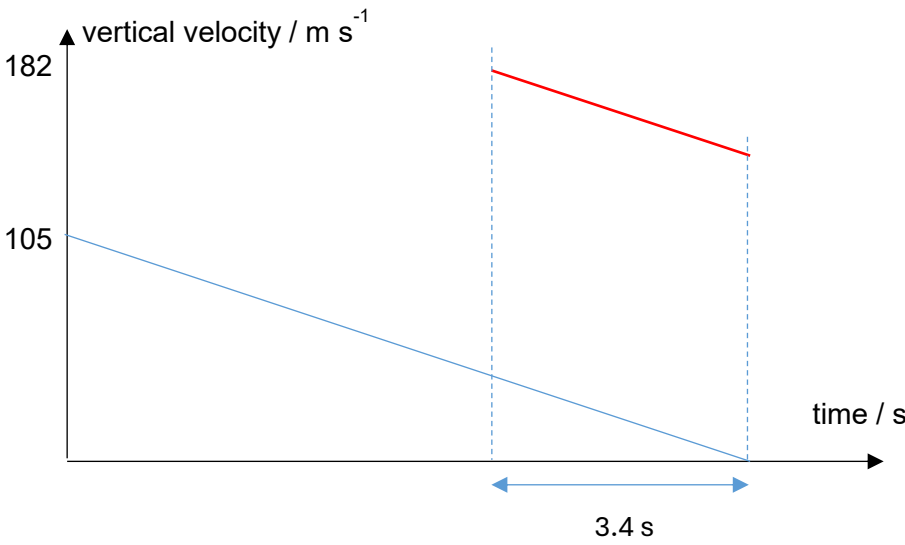


Q2		
(a)	The acceleration of an object is its rate of change of velocity with respect to time.	A1
(b)(i)	Taking upward to be positive and using $v_y^2 = u_y^2 + 2a_y s_y$, where v_y is the vertical component of the final velocity, u_y is the vertical component of the initial velocity, a_y is the vertical component of the acceleration and s_y is the vertical component of the displacement. At the highest point, the vertical component of the velocity is zero. Furthermore, ignoring air resistance, the vertical component of the acceleration is $a_y = -g = -9.81 \text{ m s}^{-2}$. Hence, $0 = (210 \sin (30^\circ))^2 + 2(-9.81) h$ $h = 561.9 \text{ m} = 560 \text{ m (1, 2 or 3 s.f.)}$	M1 A1
(b)(ii)	Taking upwards as positive and using $v_y = u_y + a_y t$ $0 = (210 \sin (30^\circ)) + (-9.81) t$ $t = 10.703 \text{ s} = 11 \text{ s (1, 2 or 3 s.f.)}$	M1 A1
(b)(iii)	From (b)(i), we know that the highest point for P is 561.9 m. Using $s_y = u_y t + \frac{1}{2} a_y t^2$, $561.9 = (210 \sin (60^\circ)) t_Q + (0.5)(-9.81) t_Q^2$ $t = 3.4 \text{ s or } t = 33.7 \text{ s}$ Since t must be less than 10.7 [from part (b)(ii)], $t = 33.7 \text{ s}$ should be rejected. $t = 3.4 \text{ s (shown)}$	B1 A0
(b)(ii)	 B1 – parallel to and above original line. (P starts at $210 \sin 30^\circ = 105 \text{ m s}^{-1}$ while Q starts at $210 \sin 60^\circ = 182 \text{ m s}^{-1}$. Values not needed in sketch) B1 – starts after mid-point of original line and ends at the same time (3.4 s not needed in sketch)	

03		
----	--	--